

Sichen (Steven) Wang

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EDUCATION

University of Toronto

Sep. 2024 – Exp. Jun 2028, Toronto

Bachelor of Engineering – Mechanical Engineering

-Relevant Courses:

MIE 243 — Mechanical Engineering Design

APS105 — C Programming

MIE 210 — Thermodynamics

MIE 230 — Vector Calculus

MIE 221 — Manufacturing Engineering

MIE 235 — Materials Science

Skill

MATLAB: Data processing, Modelling/Simulation

Excel: Data cleaning, formulas/functions

Python/C/C++: Structured programming, Debugging

CAD (SolidWorks, Fusion360): Mechanical 3D design

Minitab: Probability Modelling/Calculating

Technical Writing: Engineering reports, documentation

EXPERIENCE

Trajekt Sports

Test Engineering Intern Jun. 2026 – Present, Mississauga

- **Designed and executed structured test matrices** to characterize robotic pitching-system accuracy, repeatability, durability, and configuration sensitivity across pitch profiles, baseball types and control models.
- **Configured and operated electromechanical subsystems**, machine-vision tracking, ball-insertion mechanisms, robotic wheel assemblies, and test fixtures; controlled test variables and verified system calibration before each run.
- **Conducted 1,000+ shot endurance, wear, and performance-validation tests**, maintaining data traceability for system configurations, failure events, and accuracy trends through Asana and Excel.
- **Performed failure-mode analysis and root-cause troubleshooting** for system errors and hardware-software integration issues; translated findings into engineering feedback for robotic design verification and system improvement.

KITE Research Institute, Intern

Assistant Researcher Jul. 2025 – Sept. 2025, Toronto

- Organized **300+ experimental video datasets** and **5,000+ images** to support AI-based slip-movement detection research.
- Supported development and validation of a **camera-based computer vision pipeline** for detecting **foot-slip events from experimental video data in fall-risk research**.
- Operated designated lab equipment and followed standard procedures to support data collection workflows.
- Coordinated with different departments/teams to align requirements, share progress updates, and resolve issues.

PROJECT

Low-Cost, Long-Reach Robot Arm for Cinematography

Team Leader, CAD Technician

Sept. 2025 – Dec. 2025

- Developed a low-cost/long-reach robotic camera arm concept to create smooth, repeatable cinematic camera motion
- Balanced reach, payload, stiffness, portability, and manufacturability through structured trade-off and concept selection.
- Team lead + CAD/mechanical designer, produced layouts/models, drafted engineering specs, and drove design iterations/selection decisions
- Achieved **92% final grade (top 5% of class)** for the project deliverables and design outcomes.

Floor-integrated perturbation tile for KITE's Falls Lab

Lead Research + CAD Technician Lead

May. 2025

- Designed a concept to induce **unexpected rotational perturbations** to better analyze gait stability.
- Enabled gait-cycle-timed perturbations and planned integration with force-platform measurements for synchronized biomechanics data capture.
- **Led concept development + mechanical CAD**, translated lab needs into engineering requirements, and iterated designs with stakeholders to improve safety/feasibility

Modelling COVID-19 Spread (SIR Mathematical Model)

MAT187 Calculus 2 Course Project

Jun. 2025

- Developed an SIR (Susceptible–Infected–Recovered) differential-equation model to simulate COVID. spread dynamics.
- Implemented the model in MATLAB and ran scenario and sensitivity analyses by varying infection and recovery parameters to evaluate how they affect peak infection timing and magnitude.
- Earned **100%** for the project deliverables and analysis.

Outside of Engineering (Soft Skills)

Basketball (2023–Present): Played competitively for two years, strengthening discipline, resilience, and teamwork.

Small Sports Card Trading Business Owner (2023–Present): Operate an independent resale business by sourcing and evaluating inventory, applying market-based pricing, negotiating transactions, tracking margins, and coordinating customer fulfillment.

Rap Band Lead (2023–Present): Lead a small music group by organizing rehearsals, coordinating performances, and guiding creative direction; strengthen leadership, teamwork, and communication.

Self-Employed Tutor (2022–Present): Tutored 15+ students in mathematics, physics, and chemistry through individualized academic support while advising students on university applications.